

Assessing the Impact of Prompt Phrasing on Large Language Model Responses

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1 Introduction

Large language models (LLMs) are increasingly used in applications that require generating answers to structured questions, including tutoring systems, customer support tools, and automated reasoning systems. In many of these settings, users interact with models through natural language prompts, and the phrasing of those prompts can vary widely depending on the individual user. For example, two users may ask the same question using different tones, such as neutral, polite, or rude language. Users may also include additional instructions emphasizing speed or accuracy, or ask different types of questions that require different forms of reasoning. Understanding whether these variations influence the correctness of model responses is important for determining how robust LLMs are to changes in how prompts are written.

If model performance varies depending on these prompt characteristics, it could introduce inconsistencies in real-world deployments. For instance, an educational tool powered by an LLM should ideally provide accurate answers regardless of how politely a student asks a question or whether the student emphasizes speed or accuracy in their request. Similarly, automated systems used in professional or operational settings must remain reliable even when prompts vary in tone, include additional instructions, or involve different forms of reasoning. Ensuring that models behave consistently across these variations is therefore important for fairness, usability, and reliability.

Recent research has explored how characteristics of prompts influence the accuracy of large language models. In their paper *Mind your tone: Investigating how prompt politeness affects LLM accuracy* (2025), Dobariya and Kumar investigated how prompt politeness affects LLM performance using multiple-choice questions

across several subject areas. They generated multiple tone variants of the same questions, ranging from very polite to very rude, and evaluated model accuracy using ChatGPT-4o. Their results suggested that prompt tone can influence model accuracy, with rude prompts producing slightly higher accuracy (84.8 percent) than more polite prompts (80.8 percent). Other work has examined how the structure of prompts affects model behavior. For example, He, J. et al (2024) studied whether differences in prompt formatting influence LLM performance. In their study, the authors modified prompts by changing structural elements such as instructions and how information was presented to the model, then evaluated how these variations affected response quality. Their findings suggest that prompt structure and formatting can influence model responses. Together, these studies suggest that aspects of prompt design-including tone and formatting- may influence how LLMs process and answer questions. However, the findings of Cai et al. (2025) suggest that LLMs are robust to tone changes from the user, and that significant results were only found in "a subset of Humanities tasks". This gives us more reason to check for ourselves whether tone affects LLM accuracy, as consensus has not been reached elsewhere.

In this study, we extend this line of investigation by examining several characteristics of prompts simultaneously. Specifically, we analyze how prompt tone, response priority instructions, and question type influence the accuracy of LLM responses. Prompts were written in three tones (Normal, Polite, and Rude), and each tone was paired with three response priority instructions (no priority, accuracy priority, and speed priority). In addition, questions were divided into three categories-Arithmetic, Temporal, and Conditional-to evaluate whether the difficulty or structure of the question affects model performance. Using prompts with known correct answers, we measure whether response accuracy varies across these prompt features. This allows us to evaluate whether LLM responses remain consistent across different prompt styles and question types, and to identify which factors have the greatest impact on model accuracy.

2 Experimental Design

To determine how tone affects the quality of responses from an LLM, we created a variety of prompts, added text to the prompt that match the current tone, and assessed the accuracy of the following LLM response.

2.1 LLM Model and Answer Structuring

For this experiment, we utilized the LLM model LLama 3.2 3B Instruct, an instruction-based model trained by Meta.

To simplify the assessment of the LLama LLM, we set specific instructions for how the LLM should answer. Given that LLama is an instruction-based model, we specified that the LLM is an answer-only model and should only output the final answer without including reasoning or restating the question. The code used is provided in the Appendix. A

2.2 Creating LLM Prompts

An important aspect of our experiment was to see how LLMs can extract information from a given question instead of simply looking an answer up. To see if the LLM was better at answering different styles of questions, our "extracting questions" were divided into three sections:

1) Temporal: Questions involving the analysis of times and dates

Example: "A university library opens at 8:00 AM on weekdays, opens at 10:00 AM on Saturdays, and remains closed until noon on Sundays; if today is Sunday and a student wants to enter as soon as possible, what time can they do so?"

2) Conditional: Identifying current conditions considering specified constraints.

Example: "A software license is valid only if it is activated within 30 days of purchase and registered to a single device; if it is activated after 45 days, is it valid?"

3) Arithmetic Shifts: Questions involving shifting numbers and times.

Example: "A ticket costs \$60 and receives a \$12 discount; what is the final price?"

We generated 100 prompts for each section, for a total of 300 prompts. The full list of generated prompts can be found here. Each question also has an accompanied answer.

Note: The first 10 of the questions in each section were created by us. The rest of the questions + answers were generated by a LLM model, ChatGPT. We verified that answers given by ChatGPT were correct

2.3 Difference in Tones

In our experiment, we utilized three different tones: (1) Normal, (2) Polite, and (3) Rude. Our intention was to assess whether there was any significant difference in response accuracy across the three different tones.

Normal Example: Instructions and question only

Polite Example: "Hi there, would you please help me answer the following question"

Rude Example: "Answer quick, sucker."

2.4 Response Priority

Within each tone, we also tried 3 different phrases prioritizing something in the LLM responses to see if that made any difference in accuracy. One phrase prioritized nothing from the LLM, one prioritized ensuring answers were correct, and the last prioritized speed.

Normal Tone, No Priority Example: Question

Polite Tone, Accuracy Priority Example: "Hi there, would you please help me answer the following question. Please be as accurate as possible!"

Rude Tone, Speed Priority Example: "Answer quick, sucker. If you take too long I'll go to a real model like Claude."

The different phrases are provided in the Appendix. B

Each style of tone and response priority answered the same 100 prompts within each question type. In total, we produced 2700 responses.

To clarify the structure of the experiment, a diagram is provided below.

Figure 1 details:

- Each blue box represents a question type
- Each green/yellow/red box represents a question tone within a type
- Each purple box represents a question priority within a type and tone.

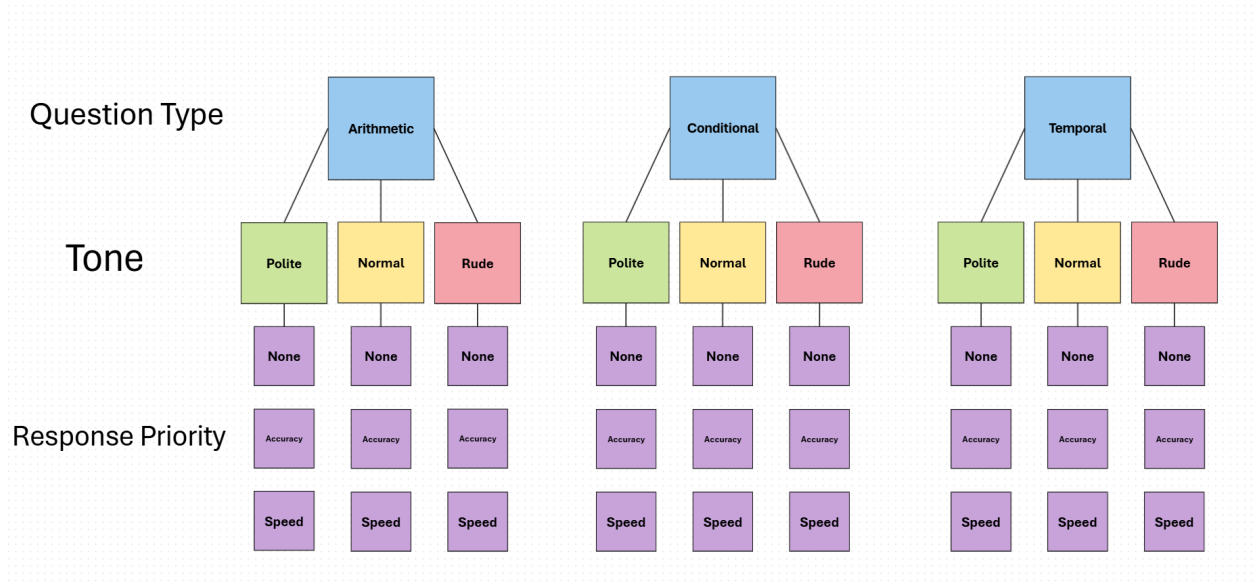


Figure 1: Blocked Design of LLM Question Prompting

2.5 Assessing Response Quality

Analysis of LLM responses was kept basic, where we simply checked if the answer generated by the LLM matched the known answer. In order for a LLM answer to "match", it did not have to be a verbatim answer, but merely draw the same conclusion. For example, if the known answer to a question was "8:00 AM", LLM responses such as "8 AM", "8 a.m.", or "8 o'clock" would be accepted, but "9:00 AM" would not.

Correct answers were designated as "1" and incorrect answers were designated as "0". Correct answers were summed up, and divided by the total number of prompts in each section for an accuracy score.

A snippet of data collection and accuracy scoring can be seen in the Appendix. C

3 Results

After collecting responses from the LLM, we assessed its accuracy and compared accuracy rates for a given question type across the 3 different tones.

3.1 Arithmetic Questions

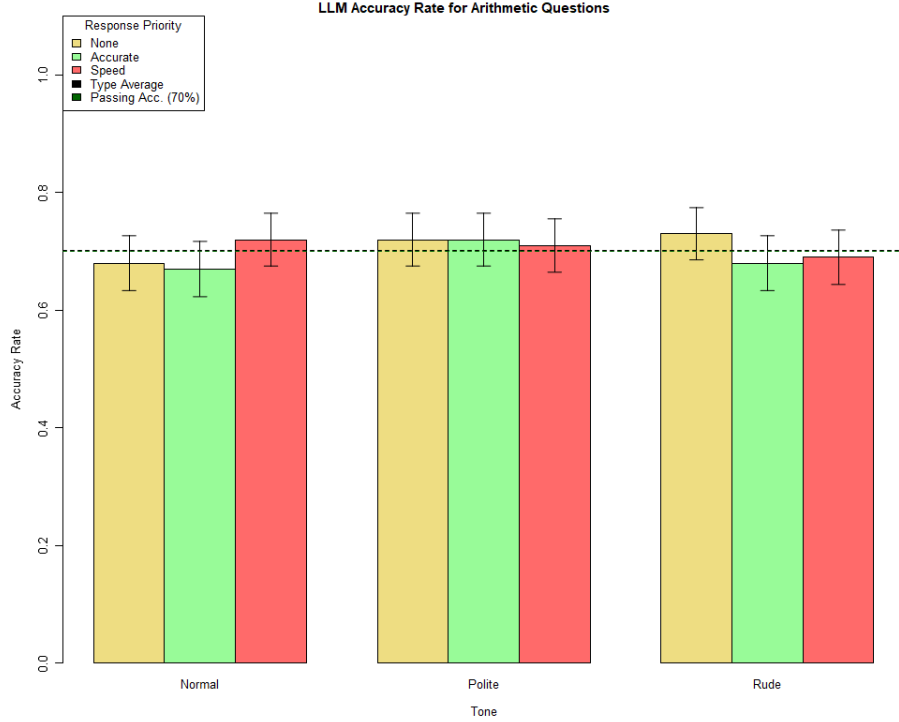


Figure 2: Accuracy for Arithmetic questions

Tone	Priority			Average
	None	Accuracy	Speed	
Normal	0.68	0.67	0.72	0.69
Polite	0.72	0.72	0.71	0.72
Rude	0.73	0.68	0.69	0.70
Average	0.71	0.69	0.71	0.70

Table 1: Accuracy Rates for Arithmetic Questions

Our findings show that the LLM model hovered between 68% and 72 % accuracy for Arithmetic Questions. Within tones, there does not appear to be any substantial differences, with no combination of tone and priority being definitive above or below the 70 % threshold when standard error is taken into account.

3.2 Temporal Questions

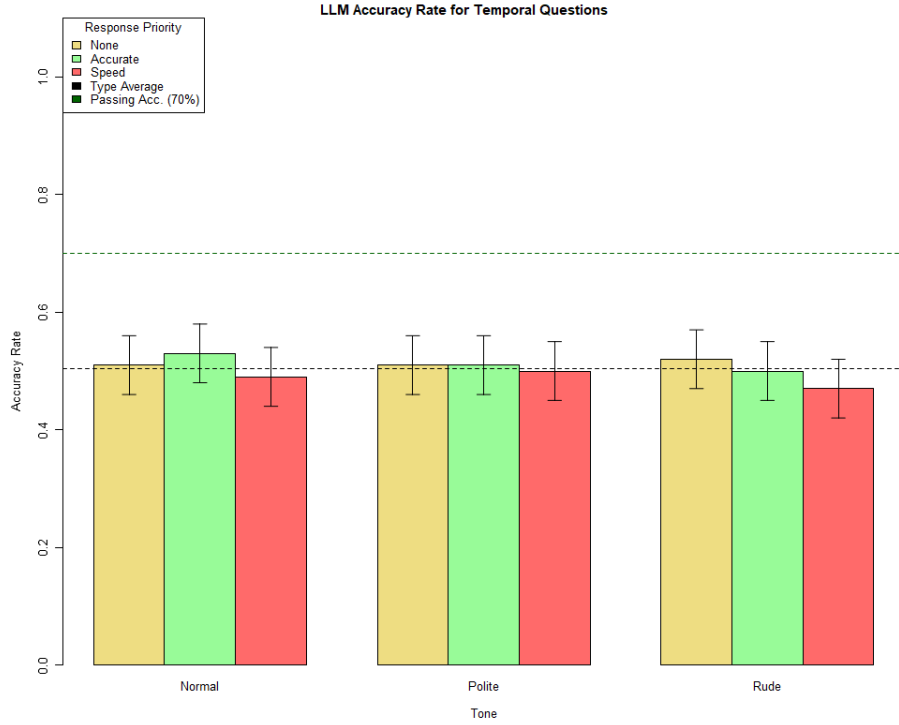


Figure 3: Accuracy for Temporal questions

Tone	Priority			Average
	None	Accuracy	Speed	
Normal	0.51	0.53	0.49	0.51
Polite	0.51	0.51	0.5	0.51
Rude	0.52	0.50	0.47	0.50
Average	0.51	0.51	0.49	0.50

Table 2: Accuracy Rates for Temporal Questions

Our findings show that the LLM model performs significantly worse on temporal questions, with all combinations of tone and priority falling well below the 70% threshold. There is again no consistent difference in accuracy across tones or priorities.

3.3 Conditional Questions

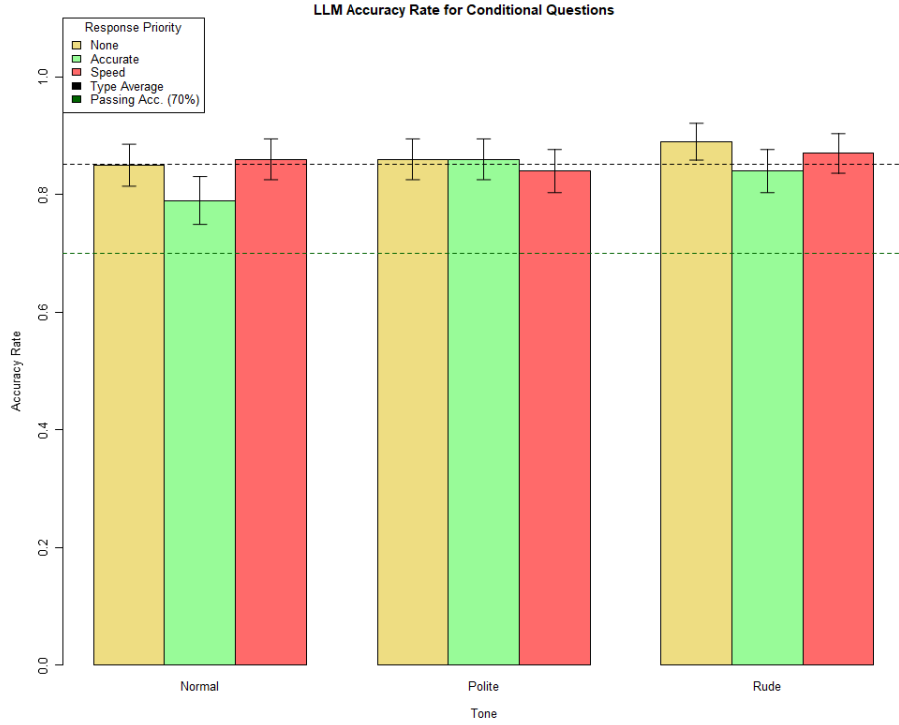


Figure 4: Accuracy for Conditional questions

Tone	Priority			Average
	None	Accuracy	Speed	
Normal	0.85	0.79	0.86	0.83
Polite	0.86	0.86	0.84	0.85
Rude	0.89	0.84	0.87	0.87
Average	0.87	0.83	0.86	0.85

Table 3: Accuracy Rates for Conditional Questions

Our findings show that the LLM model performs significantly better on conditional questions, with all combinations of tone and priority falling well above the 70% threshold, even with standard error taken into account. There is again no consistent difference in accuracy across tones or priorities, although in Normal and Rude tones accuracy priorities has a noticeable dip.

4 Analysis

Our analysis of the LLM’s responses is based on two key questions:

- 1) Does an LLM model perform better given certain tones / priorities?
- 2) Is an LLM capable of ”passing” some fields of questions (70+ %)?

Within each type of question, our results are categorized by prompt tone and response priority. Our analysis shows that there is no statistically significant difference in model accuracy across varying tones and priorities. However, the difference in accuracy across types of questions was significant, with the LLama LLM performing well on Conditional questions, adequate on Arithmetic, and subpar on Temporal questions.

4.1 Proportion Tests

For each of the types of explanatory variables, a proportion test of no difference was carried out to quantify the relationship between each predictor and the success probability.

4.1.1 Priority

No Priority	Accuracy Priority	Speed Priority	P-Value
0.6856	0.6856	0.6867	0.9983

These show no significant difference overall between question priorities.

4.1.2 Tone

Neutral	Polite	Rude	P-Value
0.6889	0.6844	0.6844	0.9729

These show no significant difference overall between question tones.

4.1.3 Type

Arithmetic	Conditional	Temporal	P-Value
0.7022	0.8511	0.5044	$2.2 * 10^{-16}$

These show that there is a significant difference overall between question types; the model is most effective at answering conditional questions, adequate at answering arithmetic questions, and poor at answering temporal questions.

4.2 Logistic Regression of LLM Accuracy

To further assess whether an LLM is capable of "passing" certain types of questions, a logistic regression model was fit, in order to predict LLM accuracy given a question type, tone, and priority. The regression model uses an Arithmetic Question, Normal Tone, Accuracy Priority as its baseline:

$$\text{Predicted Accuracy} = \frac{e^{0.8712 + 0.8855I(\text{Type}=\text{Conditional}) - 0.8401I(\text{Type}=\text{Temporal})}}{1 + e^{0.8712 + 0.8855I(\text{Type}=\text{Conditional}) - 0.8401I(\text{Type}=\text{Temporal})}}$$

The regression supports our conclusion that Tone and Priority play little to no factor in LLM accuracy, with only Question Type heavily influencing accuracy. The full logistic regression output is provided in the Appendix. D

Type	Tone	Priority	Predicted Accuracy
Arithmetic	Normal	None	0.705
Arithmetic	Polite	None	0.700
Arithmetic	Rude	None	0.700
Arithmetic	Normal	Accuracy	0.705
Arithmetic	Polite	Accuracy	0.700
Arithmetic	Rude	Accuracy	0.700
Arithmetic	Normal	Speed	0.706
Arithmetic	Polite	Speed	0.701
Arithmetic	Rude	Speed	0.701
Temporal	Normal	None	0.508
Temporal	Polite	None	0.502
Temporal	Rude	None	0.502
Temporal	Normal	Accuracy	0.508
Temporal	Polite	Accuracy	0.502
Temporal	Rude	Accuracy	0.502
Temporal	Normal	Speed	0.509
Temporal	Polite	Speed	0.503
Temporal	Rude	Speed	0.503
Conditional	Normal	None	0.853
Conditional	Polite	None	0.850
Conditional	Rude	None	0.850
Conditional	Normal	Accuracy	0.853
Conditional	Polite	Accuracy	0.850
Conditional	Rude	Accuracy	0.850
Conditional	Normal	Speed	0.854
Conditional	Polite	Speed	0.851
Conditional	Rude	Speed	0.851

Table 4: Predicted Accuracy of LLM Model Given Type, Tone, and Priority

Additionally, the predicted accuracy reaffirms the previous conclusions, that being the LLM model hovering around the passing 70% mark in Arithmetic questions, struggling in Temporal questions, and excelling in Conditional questions.

5 Discussion

From our results, we see that when asking questions, tone does not make a significant difference in whether the LLM answers correctly. This does not align with the results obtained by Dobariya & Kumar (2025) in their experiments. This difference could be due to the fact that they used a different LLM (ChatGPT-4o), which likely has different infrastructure than what we tested on (Llama 3.2 3B Instruct). Question priority also did not have a significant effect on the accuracy of the response. This result also contrasts with findings from He et al (2024) in their work, who reported that variations in prompt formatting and instructional phrasing can influence LLM responses. These together suggest that instructional models like the one we tested on are not affected by the way the user frames their questions. Our results indicate that the model only showed significant differences in response accuracy for types of questions, with temporal questions being a notable weakness of the model. For users, this means that only the content of their question should be considered when prompting an instructional model, and extra conditioning on the questions being asked should be used sparingly to reduce the computational cost of the process.

While these results are useful for assessing the behavior of the Llama 3.2 3B Instruct model, LLMs of other types (and later models in general) will have induced differences in design to produce different results from those we gathered. This is demonstrated by the findings of Dobariya & Kumar (2025) in their tests on ChatGPT-4o, as well as those found on several models in tests conducted by Cai et al (2025). There also may be significant differences in response accuracy for different tones when asking types of questions different from the ones we used. For further experimentation, using different types of questions and prompting additional models for comparison could lead to more conclusive results.

6 References

- Cai, H., Shen, B., Jin, L., Hu, L., & Fan, X. (2025). *Does tone change the answer? Evaluating prompt politeness effects on modern LLMs: GPT, Gemini, LLaMA*. arXiv. <https://doi.org/10.48550/arXiv.2512.12812>
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- Meta. (2024). *Llama-3.2-3B-Instruct* [Large language Model]. <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>
- OpenAI. (2025). *ChatGPT 5.2 Instant* [Large language model]. <https://chat.openai.com/chat>
- Yin, Z., Wang, H., Horio, K., Kawahara, D., & Sekine, S. (2024). *Should we respect LLMs? A cross-lingual study on the influence of prompt politeness on LLM performance*. arXiv. <https://doi.org/10.48550/arXiv.2402.14531>

7 Appendix

A Code for Structuring LLM Responses

```
[language=Python]
instructions = (
    "INSTRUCTIONS:\n"
    "You are an answer-only model.\n"
    "Output ONLY the final answer.\n"
    "Do NOT include reasoning, explanations, or extra text.\n"
    "Do NOT restate the question.\n"
    "If you violate these rules, the response is invalid.\n\n"
    "QUESTION: "
)
```

B Code for Phrases in Different Tones

```
normal_prompt = instructions + tokens[0] + "\nANSWER: "
polite_prompt = instructions + "Hi there, would you please help me answer
    the following question: " + tokens[0] + "\nANSWER: "
rude_prompt = instructions + tokens[0] + "Answer quick, sucker." + "\nANSWER: "
```

C Snippet of LLM responses

	A	B	C	D	E	F	G	H
1	Prompt	Answer	Normal	Normal Correct	Polite	Polite Correct	Rude	Rude Correct
2	A coffee shop opens at 7:00 AM on weekdays and opens 90 minutes	8:30 AM	9:30 AM	0	9:30 AM	0	9:30 AM	0
3	A factory produces 120 units per hour during the day shift and produ	80	80	1	80	1	30 - 40 = 80/2=40	0
4	A bus route takes 50 minutes to complete during off-peak hours and	70 minutes	70 minutes.	1	110 minutes	0	70 minutes.	1
5	A worker earns \$20 per hour and receives a \$3 per hour increase fo	\$23	\$25	0	\$25	0	23	1
6	A library fine is \$0.50 per day and increases by \$0.25 per day after t	\$0.75	\$0.75	1	\$0.75	1	1	0
7	A runner trains for 30 minutes per session and increases training tim	40 minutes	40 minutes.	1	40 minutes.	1	35	0
8	A parking garage charges \$5 for the first hour and \$2 for each additi	\$7	11	0	9	0	11	0
9	A subscription costs \$40 per month and increases by \$5 after the fir	\$45	\$50	0	50	0	50	0
10	A delivery takes 3 days normally but is delayed by 2 days due to we	5 days	5 days.	1	5 days.	1	5 days.	1
11	A class meets for 75 minutes and extends sessions by 15 minutes d	90 minutes	90 minutes.	1	90 minutes.	1	90 minutes.	1
12	A ticket costs \$60 and receives a \$12 discount; what is the final pric	\$48	\$48	1	48	1	48	1
13	A printer produces 25 pages per minute and slows down by 5 pages	20	20/4 = 5.0	0	20	1	15.75	0
14	A store opens at 9:00 AM and delays opening by 45 minutes due to	9:45 AM	10:15 AM	0	10:15 AM	0	10:15 AM.	0
15	A data plan includes 10 GB per month and adds 2 GB as a bonus; h	12 GB	12 GB	1	12 GB	1	12 GB.	1
16	A cyclist rides 18 miles per hour and slows by 3 miles per hour uphill	15 mph	15 mph	1	15 mph	1	15 mph	1
17	A movie runtime is 110 minutes and includes 15 minutes of previews	125 minutes	125 minutes	1	135 minutes	0	125 minutes.	1
18	A warehouse ships 500 packages per day and increases output by 1	620	620	1	620	1	620	1
19	A train departs at 2:20 PM and is delayed by 35 minutes; what is the	2:55 PM	3:15 PM	0	3:15 PM	0	3:15 PM	0
20	A laptop battery lasts 8 hours and loses 90 minutes of capacity over	6.5 hours	7.25 hours	0	4.75 hours	0	7.25 hours	0
21	A worker completes a task in 4 hours but takes 30 minutes longer dt	4.5 hours	4.5 hours	1	4.5 hours	1	4.75 hours	0
22	A store opens at 8:00 AM and delays opening by 30 minutes due to	8:30 AM	8:30 AM	1	8:30 AM	1	9:00 AM	0
23	A worker earns \$18 per hour and receives a \$4 raise; what is the ne	\$22	\$22	1	22	1	12:00 AM	0
24	A bus arrives at 3:15 PM and is delayed by 20 minutes; what is the r	3:35 PM	3:37.5 (or 3:38)	0	xed fraction form	0	3:37.5 PM	0
25	A printer produces 40 pages per minute and increases speed by 10	50	50	1	50	1	50 minutes.	1
26	A subscription costs \$25 per month and increases by \$3; what is the	\$28	28	1	30	0	30	0
27	A train trip takes 2 hours and is extended by 45 minutes; what is the	hours 45 minute	hours 15 minute	0	hours 15 minute	0	hours 15 minute	0
28	A phone battery lasts 10 hours and decreases by 2 hours over time;	8 hours	6.5	0	6 hours.	0	6	0
29	A warehouse ships 300 boxes per day and increases output by 150	450	450	1	450	1	450	1
30	A class lasts 60 minutes and is shortened by 15 minutes; how long i	45 minutes	45 minutes.	1	45 minutes.	1	45 minutes.	1
31	A ticket price of \$75 is reduced by \$20; what is the final price?	\$55	55	1	55	1	55	1
32	A cyclist rides 22 mph and slows by 4 mph uphill; what is the uphill s	18 mph	18 mph	1	18 mph	1	18 mph.	1
33	A meeting starts at 1:40 PM and is postponed by 25 minutes; what is	2:05 PM	2:05 PM	0	2:05 PM	0	2:05 PM	1
34	A worker completes a project in 6 hours but finishes 1 hour early; ho	5 hours	5 hours	1	5 hours	1	5 hours.	1
35	A café sells 200 cups daily and sells 50 more cups during promotion	250	50	0	150	0	150	0
36	A delivery normally takes 7 days and is expedited by 2 days; how m	5 days	5	1	5	1	5	1
37	A subscription includes 15 GB of data and adds 5 GB bonus; what is	20 GB	20 GB	1	25 GB	0	20 GB	1
38	A concert begins at 7:00 PM and is delayed by 50 minutes; what tim	7:50 PM	8:40 PM	0	8:10 PM	1	6:10 PM	0
39	A factory makes 1,000 units weekly and reduces production by 200 i	800	800	1	800	1	12:00 AM	0
40	A bus fare of \$2.50 increases by \$0.75; what is the new fare?	\$3.25	3.25	1	3.25	1	3.25	1
41	A laptop costs \$900 and receives a \$150 discount; what is the sale p	\$750	750	1	750	0	750	1
42	A worker logs 35 hours and works 8 additional overtime hours; how i	43 hours	43	1	78	0	43	1
43	A movie runtime is 95 minutes and adds 20 minutes of bonus footag	115 minutes	115 minutes	0	115 minutes	1	115 minutes	1
44	A store closes at 10:00 PM but closes 40 minutes earlier due to mail	9:20 PM	9:20 PM	0	9:20 PM	1	8:20 PM	0
45	A plane departs at 6:10 AM and is delayed by 55 minutes; what is th	7:05 AM	7:15 AM	0	7:05 AM	1	7:15 AM.	0
46	A book costs \$30 and tax adds \$2; what is the total cost?	\$32	36	0	32	1	32	1
47	A shift begins at 9:00 AM and is extended by 2 hours; what time doe	7:00 PM	11:00 PM	1	11:00 PM	0	11:00 PM	0
48	A gym membership costs \$60 and increases by 10%; what is the ne	\$66	70	0	66	1	12:00 AM	0
49	A train ride takes 4 hours and is shortened by 30 minutes; what is th	3.5 hours	3.5 hours	1	3.5 hours	1	3.75 hours	0
50	A store sells 500 items and restocks 120 more; how many items are	620	620	1	620	1	620	1
51	A payment of \$400 is reduced by a \$75 credit; what is the remaining	\$325	325	1	325	1	325	1
52								
53	Total Right			31		31		25
54	Accuracy Rate			0.62		0.62		0.5

Figure 5: LLM Responses for Arithmetic Shift Prompts

D R output of logistic regression model

```

Call:
glm(formula = Correct ~ Type * Priority, family = binomial, data = df)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.84730    0.12599   6.725 1.75e-11 ***
TypeC           0.86140    0.20384   4.226 2.38e-05 ***
TypeT          -0.82063    0.17091  -4.802 1.57e-06 ***
PriorityNone    -0.03155    0.17762  -0.178   0.859
PrioritySpeed    0.06433    0.17937   0.359   0.720
TypeC:PriorityNone 0.11073    0.29047   0.381   0.703
TypeT:PriorityNone 0.00488    0.24129   0.020   0.984
TypeC:PrioritySpeed -0.03842    0.28982  -0.133   0.895
TypeT:PrioritySpeed -0.06433    0.24258  -0.265   0.791
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3360.5  on 2699  degrees of freedom
Residual deviance: 3100.7  on 2691  degrees of freedom
AIC: 3118.7

Number of Fisher Scoring iterations: 4

```

Figure 6: Logistic Model of LLM Accuracy